# scholars.ahsansuny.com

Free Deployment Workplan

| $0/month Phase 1 & 2 | 3 Phases 8 weeks | Claude API Extraction + RAG | Vercel + Supabase + Pinecone free |
|---|---|---|---|

## 1. What You Already Have

| Asset | Details | Role |
|---|---|---|
| ahsansuny.com | Your own domain | scholars.ahsansuny.com via CNAME → Vercel. Free SSL auto-provisioned. |
| Claude API | Anthropic API key | Haiku for batch extraction + classification. Sonnet for student RAG queries. |
| Claude Code API | Agentic coding | Your local dev agent — writes scrapers, Flask/Next.js routes, DB queries for you. |

## 2. Complete Free Stack

| Layer | Tool | Free Limit | Purpose |
|---|---|---|---|
| Frontend | Vercel Free | Unlimited | Next.js app on scholars.ahsansuny.com. CNAME → Vercel. Auto SSL. |
| Backend API | Vercel API Routes | 1M calls/mo | Next.js API routes handle all requests. No separate server needed. |
| Database | Supabase Free | 500 MB / 50k rows | PostgreSQL for opportunities, users, bookmarks, alerts. Auth included. |
| Vector Search | Pinecone Free | 100k vectors | Semantic search over all opportunities. 100k = Phase 1 + 2 fully covered. |
| AI Extraction | Claude Haiku | ~$0.40/mo | Batch: title, type, deadline, nationality, field from raw scraped text. |
| AI RAG Search | Claude Sonnet | ~$1–3/mo | Student queries: "PhD in Germany for Pakistani students". Conversational. |
| Embeddings | HuggingFace Free | Unlimited | all-MiniLM-L6-v2 via Inference API. Free, fast, good quality. |
| Auth | Supabase Auth | 50k MAU | Email + Google OAuth. Built into Supabase free tier. |
| Crawling | GitHub Actions | 2000 min/mo | Scheduled cron workflows run scrapers daily. Free on public repos. |
| Email | Brevo Free | 300 emails/day | Weekly digest + deadline alerts for student community. |
| Telegram Bot | Telegram API | Free forever | /search /today /deadline commands. Zero cost at all. |
| Domain | ahsansuny.com ✓ | Already owned | scholars.ahsansuny.com — one CNAME record in your registrar. |

| | | | |
|---|---|---|---|
| **Total monthly cost (Phase 1 & 2)** | **$0 on all free tools** | **+ ~$3–4 Claude API** | **= ~$3–4 / month total** |

## 3. Phased Workplan

### Phase 1 — Student MVP

Weeks 1–3 | $0

| | |
|---|---|
| **DNS Setup** | Add CNAME record: scholars → cname.vercel-dns.com in your registrar. scholars.ahsansuny.com goes live with SSL. |
| **Supabase** | Create free project. Run schema SQL: opportunities, users, bookmarks tables. Get API keys. |
| **Pinecone** | Create free index (768 dims, cosine). Get API key for embedding storage. |
| **GitHub Actions** | Set up 3 daily scrapers: EURAXESS (Playwright), DAAD, OpportunityDesk. Store secrets in repo. |
| **Claude Pipeline** | GitHub Action calls Claude Haiku to extract + classify each scraped opportunity. Embeds with HuggingFace → Pinecone. |
| **Next.js App** | Deploy to Vercel. Opportunity list page + detail page. Basic keyword search using Supabase full-text. |
| **Goal** | scholars.ahsansuny.com is live with 200+ real opportunities. Share with your students. |

### Phase 2 — Smart Features

Weeks 4–7 | $0

| | |
|---|---|
| **Student Auth** | Enable Supabase Auth. Add Google OAuth. Student profile: nationality, field, degree level. |
| **Personalised Feed** | After login, filter feed by student nationality + field. Show only eligible opportunities. |
| **RAG Search** | Build /api/search route. Hybrid: Pinecone vector query + Supabase keyword. Claude Sonnet generates ranked answer. |
| **Chat UI** | Conversational search box on homepage. Student types "PhD in AI for Pakistani students" → ranked results + explanation. |
| **Bookmarks & Alerts** | Save opportunities. Cron job checks deadlines, sends alerts 30/14/7 days before via Brevo email. |
| **Telegram Bot** | python-telegram-bot deployed to Vercel or free Railway. /search, /today, /deadline commands. |
| **More Scrapers** | 5 more: FindAPhD, Nature Careers, jobs.ac.uk, Chevening, Commonwealth Scholarships. |
| **Goal** | 500+ opportunities, student login, personalised feed, working RAG chat, Telegram bot live. |

### Phase 3 — Community Growth

Weeks 8–12 | Still $0

| | |
|---|---|
| **Community Submit** | Public form to submit new opportunities. Claude Haiku validates + classifies before publishing. |
| **Telegram Ingest** | Bot monitors 10+ public scholarship channels. Auto-extracts + adds to database. |
| **App Assistant** | RAG-powered chat: "What documents do I need for Chevening?" — Claude Sonnet answers from opportunity data. |
| **PWA** | Add next-pwa. Students install on phone. Push notifications for deadline alerts. |
| **More Scrapers** | Reach 5,000+ opportunities: UKRI, NWO, Swiss Gov, MEXT Japan, GKS Korea, Türkiye Burslari. |
| **SEO Pages** | Static SSG pages: "Scholarships for Pakistani students", "PhD positions in Germany". Google traffic. |
| **Upgrade Signal** | When Supabase hits 500MB → upgrade to $25/mo Pro. By then you have users to justify it. |

## 4. Supabase Database Schema (SQL)

Run this SQL in the Supabase SQL editor to create all tables

```
-- Core table
CREATE TABLE opportunities (
  id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title            TEXT NOT NULL,
  type             TEXT,          -- scholarship|grant|phd|postdoc|internship|fellowship
  host_country     TEXT[],        -- ['DE','UK']
  eligible_nations TEXT[],        -- ['ALL'] or ['PK','BD'] or ['DEVELOPING']
  field_of_study   TEXT[],
  degree_level     TEXT[],        -- bachelor|master|phd|postdoc|any
  funding_type     TEXT,          -- full|partial|stipend|salary
  amount_usd       INTEGER,
  deadline         DATE,
  status           TEXT DEFAULT 'open',   -- open|closed|rolling
  description      TEXT,
  apply_url        TEXT,
  source_url       TEXT NOT NULL,
  source_name      TEXT,          -- DAAD|EURAXESS|LinkedIn|...
  embedding_id     TEXT,          -- Pinecone record ID
  created_at       TIMESTAMPTZ DEFAULT now(),
  updated_at       TIMESTAMPTZ DEFAULT now()
);

-- Student profiles (linked to Supabase Auth user)
CREATE TABLE profiles (
  id               UUID PRIMARY KEY REFERENCES auth.users(id),
  full_name        TEXT,
  nationality      TEXT[],        -- ['PK','GB']
  field_of_study   TEXT[],
  degree_level     TEXT,
  target_countries TEXT[],
  digest_freq      TEXT DEFAULT 'weekly',   -- daily|weekly|never
  telegram_id      TEXT,
  created_at       TIMESTAMPTZ DEFAULT now()
);

-- Bookmarks
CREATE TABLE bookmarks (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id       UUID REFERENCES profiles(id),
  opp_id        UUID REFERENCES opportunities(id),
  notes         TEXT,
  created_at    TIMESTAMPTZ DEFAULT now(),
  UNIQUE(user_id, opp_id)
);

-- Useful indexes
CREATE INDEX idx_opp_deadline ON opportunities(deadline);
CREATE INDEX idx_opp_status   ON opportunities(status);
CREATE INDEX idx_opp_type     ON opportunities(type);
CREATE INDEX idx_opp_nations  ON opportunities USING GIN(eligible_nations);
```

## 5. Claude API — How to Use It

| Task | Model | When Called | Est. Cost |
|---|---|---|---|
| Opportunity extraction | claude-haiku-4-5 | On each new scraped item (batch) | ~$0.40/mo |
| Nationality classification | claude-haiku-4-5 | Alongside extraction | ~$0.30/mo |
| RAG search answer | claude-sonnet-4-6 | Per student query (on-demand) | ~$1.50/mo |
| Application assistant | claude-sonnet-4-6 | Per student session (on-demand) | ~$1.00/mo |
| Digest personalisation | claude-haiku-4-5 | Weekly, per student | ~$0.20/mo |

| | | | |
|---|---|---|---|
| **TOTAL (100 students)** | | | **~$3.40/mo** |

**Extraction prompt skeleton (Haiku):**

```
SYSTEM: Extract scholarship data. Return ONLY JSON.
Fields: title, type, host_country[], eligible_nations[], field_of_study[],
        degree_level[], funding_type, amount_usd, deadline (YYYY-MM-DD), apply_url
Rules: eligible_nations = ["ALL"] if open to everyone.
       Use ISO-2 country codes. Return null for unknown fields.

USER: {raw_scraped_text}
```

**RAG search prompt skeleton (Sonnet):**

```
SYSTEM: You are ScholarPath. Given retrieved opportunities, rank by relevance
        to the user query and profile. For each: explain the match (2 sentences),
        highlight deadline + funding + eligibility. Never invent data.

USER: Query: {user_query}
      Nationality: {nationality} | Field: {field} | Level: {level}

      Retrieved opportunities:
      {context}
```

## 6. Repository Structure

```
scholars-platform/
■■■ app/                            # Next.js 14 app
■   ■■■ page.tsx                    # Homepage + search
■   ■■■ opportunities/[id]/         # Detail page
■   ■■■ dashboard/                  # Student dashboard
■   ■■■ api/
■       ■■■ search/route.ts         # RAG search endpoint
■       ■■■ opportunities/          # CRUD endpoints
■       ■■■ alerts/                 # Deadline alerts cron
■■■ crawlers/                       # Python scrapers
■   ■■■ base.py                     # BaseCrawler class
■   ■■■ euraxess.py                 # Playwright scraper
■   ■■■ daad.py
■   ■■■ opportunitydesk.py
■   ■■■ pipeline.py                 # Claude extract → embed → upsert
■■■ .github/workflows/
■   ■■■ crawl.yml                   # Daily cron: runs all scrapers
■■■ lib/
■   ■■■ supabase.ts                 # DB client
■   ■■■ pinecone.ts                 # Vector search client
■   ■■■ claude.ts                   # Claude API wrapper
■■■ .env.local                      # API keys (never commit)
■■■ README.md
```

## 7. Environment Variables (.env.local)

```
# Supabase
NEXT_PUBLIC_SUPABASE_URL=https://xxxx.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJ...
SUPABASE_SERVICE_KEY=eyJ...          # server-side only

# Pinecone
PINECONE_API_KEY=pcsk_...
PINECONE_INDEX=scholars-opps

# Claude (Anthropic)
ANTHROPIC_API_KEY=sk-ant-...
```

```
# HuggingFace (embeddings)
HF_API_TOKEN=hf_...

# Brevo (email)
BREVO_API_KEY=xkeysib-...

# Telegram Bot
TELEGRAM_BOT_TOKEN=123456:ABC-...
```

## 8. Week 1 — Day-by-Day Checklist

**Day 1**

1. Create GitHub repo: scholars-platform (public)
2. Sign up: Supabase free, Pinecone free, Vercel free, Brevo free, HuggingFace
3. Run schema SQL in Supabase → all tables created
4. Deploy empty Next.js app to Vercel → connect scholars.ahsansuny.com via CNAME

**Day 2**

1. Write crawlers/base.py (BaseCrawler class)
2. Write crawlers/euraxess.py (Playwright scraper, ~50 opportunities)
3. Test: run scraper locally → 50 records in Supabase ✓

**Day 3**

1. Write crawlers/pipeline.py: calls Claude Haiku → extracts JSON → embeds with HF → upserts to Pinecone + Supabase
2. Test end-to-end: raw HTML → structured row in DB + vector in Pinecone ✓

**Day 4**

1. Add DAAD + OpportunityDesk scrapers
2. Run all 3 scrapers → 200+ opportunities in DB
3. Build Next.js opportunity list page (cards with title, type, deadline, country)
4. Build opportunity detail page with apply button

**Day 5**

1. Set up .github/workflows/crawl.yml — daily cron that runs all scrapers
2. Push to GitHub → verify GitHub Actions runs successfully
3. Share scholars.ahsansuny.com URL with your students ■
4. Collect feedback — what do they search for most?

## 9. Upgrade Path — When & What to Pay For

| Trigger | Upgrade | From → To | Cost |
| --- | --- | --- | --- |

| | | | |
|---|---|---|---|
| DB hits 500MB | Database | Supabase Free → Pro | $25/mo |
| 100k+ opportunities | Vector search | Pinecone Free → Starter | $70/mo |
| 300+ emails/day | Email service | Brevo Free → Starter | $9/mo |
| 500+ daily users | Backend | Vercel Free → Pro or VPS | $20/mo |
| Need always-on bot | Bot hosting | Vercel cron → Railway free | $0 |
| 26 Pro users at $5/mo | | | **Covers ALL upgrades above** |

**scholars.ahsansuny.com**

Built with Claude API · Vercel · Supabase · Pinecone · GitHub Actions